

HAT-P-19b

R-1328

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_231005 · created 2026-07-31T20:17:39+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460222.7861 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.013
Transit depth (Rp/Rs)^2	2.37 +/- 0.4 [%]
Orbital Inclination (inc)	90.0 +/- 0.5
Airmass coefficient 1 (a1)	1.12 +/- 0.014
Airmass coefficient 2 (a2)	-0.1 +/- 0.032
Scatter in the residuals of the lightcurve fit is	1.26 %
Best Comparison Star	#3 - [372, 265]
Optimal Aperture	3.59
Optimal Annulus	9.4
Transit Duration (day)	0.1223 +/- 0.0016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.154 $\pm$ 0.013 vs 0.1418 (deviation 0.78 σ)
Duration fit vs published	0.1223 d vs 0.1196 d (deviation 1.41 σ)
Mid-time O-C	12.1 min
Depth SNR	9.9
Residual scatter	1.26 %

Light curve

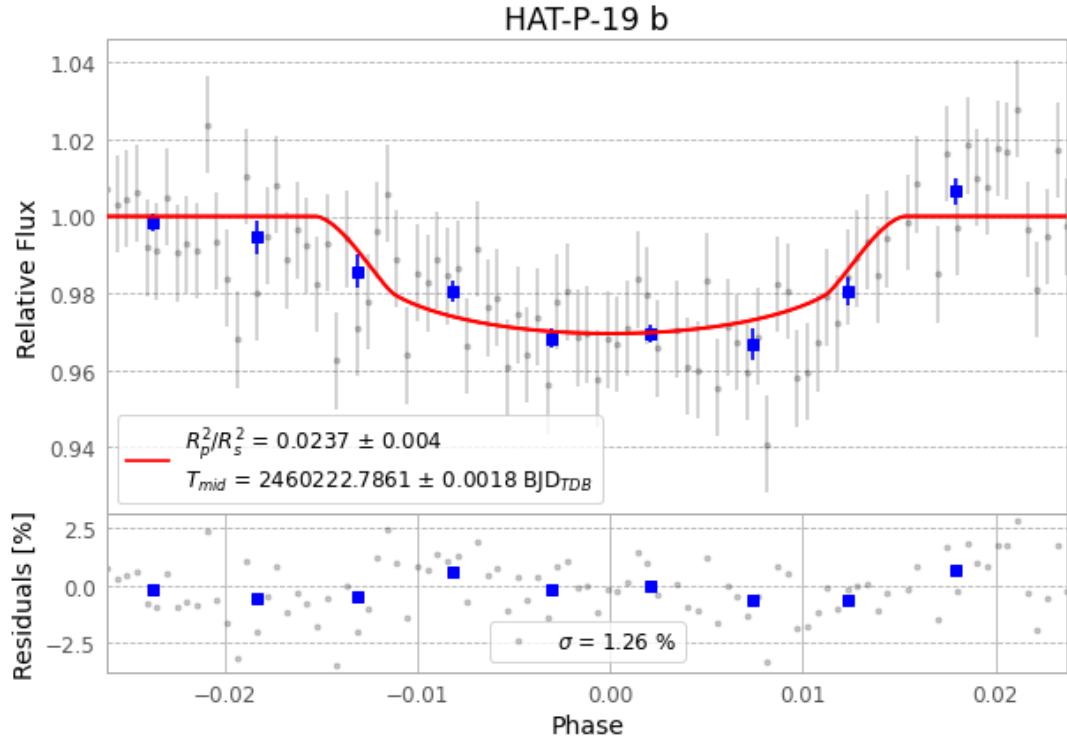


Figure 1 — FinalLightCurve_HAT-P-19 b_2023-10-04.png (SHA-256 57aecf94bb9dda2e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [294, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f6027f12f02fc9e7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

97 FITS frames (64 MB), HATP-19231005041215.FITS → HATP-19231005090015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-231005.log (01da906b8684...), FinalLightCurve_HAT-P-19 b_2023-10-04.png (57aecf94bb9d...), FinalLightCurve_HAT-P-19 b_2023-10-04.csv (871d57f291a4...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 20:17Z.